

MECHTRON 2MD3: Assignment 3

PART 1

Part 1.1:

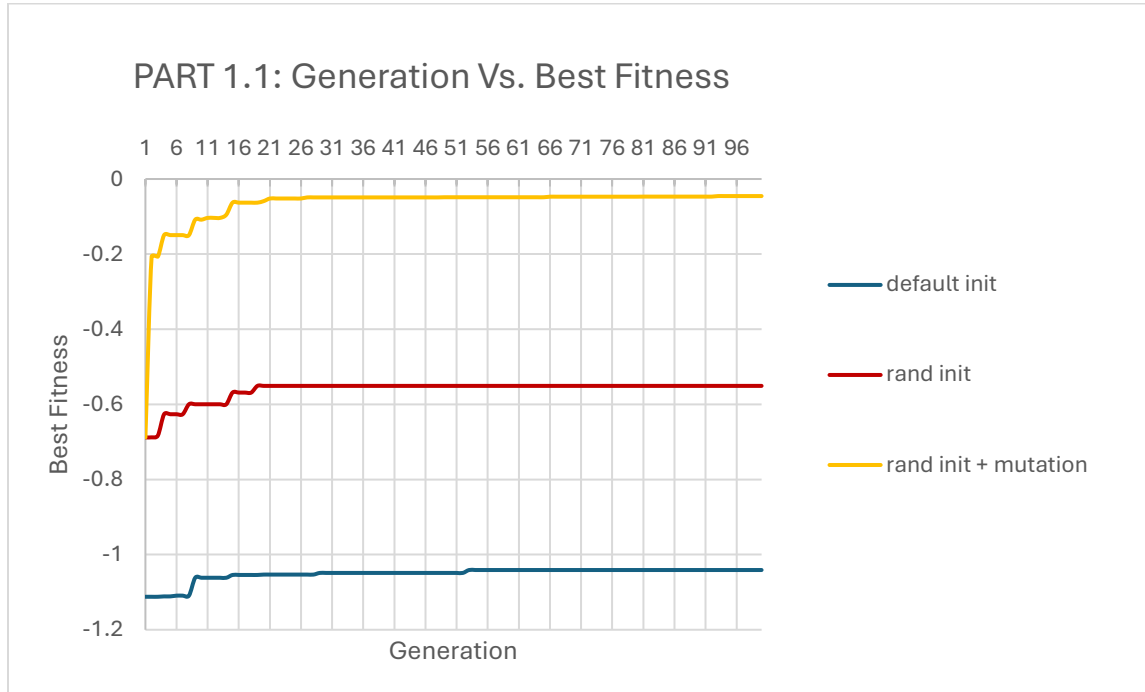


Figure 1: Graph displaying values determined in Part 1 for Genetic Programming.

Part 1.2:

Best tree:

$$(a/((((b-(a*abs((((b*a)/(abs(b)+(a>(b/(b*a))))))>(a*((a>a)>((b-a)*abs(b)))))))*((a>b)*b))>a)+(((a-(a/b))+b)*b)/((a/abs(b))+a)))$$

Generation: 92

Size: 59

Depth: 14

Fitness: -0.0454425

PART 2

Part 2.1:

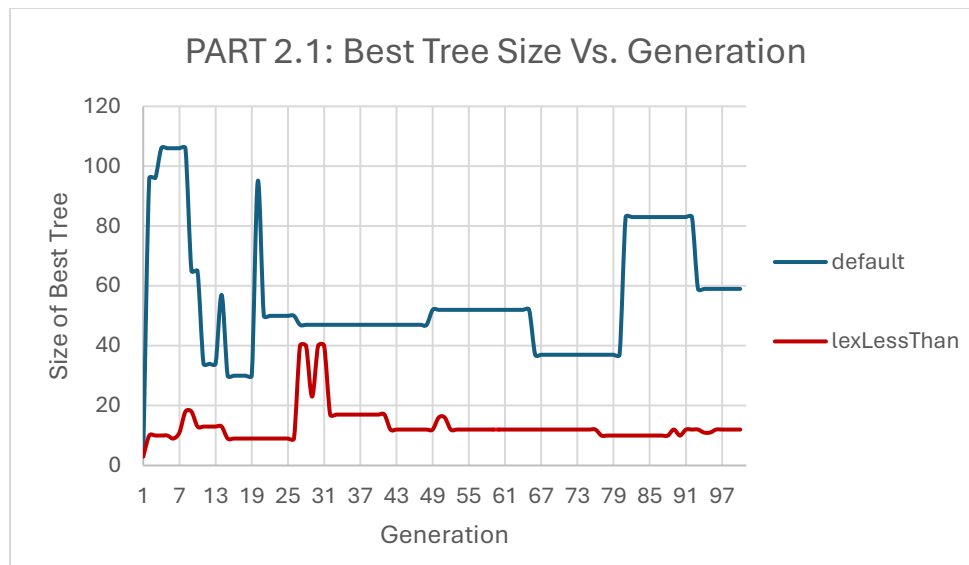


Figure 2: Graph displaying values determined in Part 2 for Genetic Programming.

Part 2.2:

Best tree:

$((b * (b * a)) - (b * \text{abs}(b))) - a$

Generation: 64

Size: 12

Depth: 4

Fitness: -0.0533381

PART 3

Part 3:

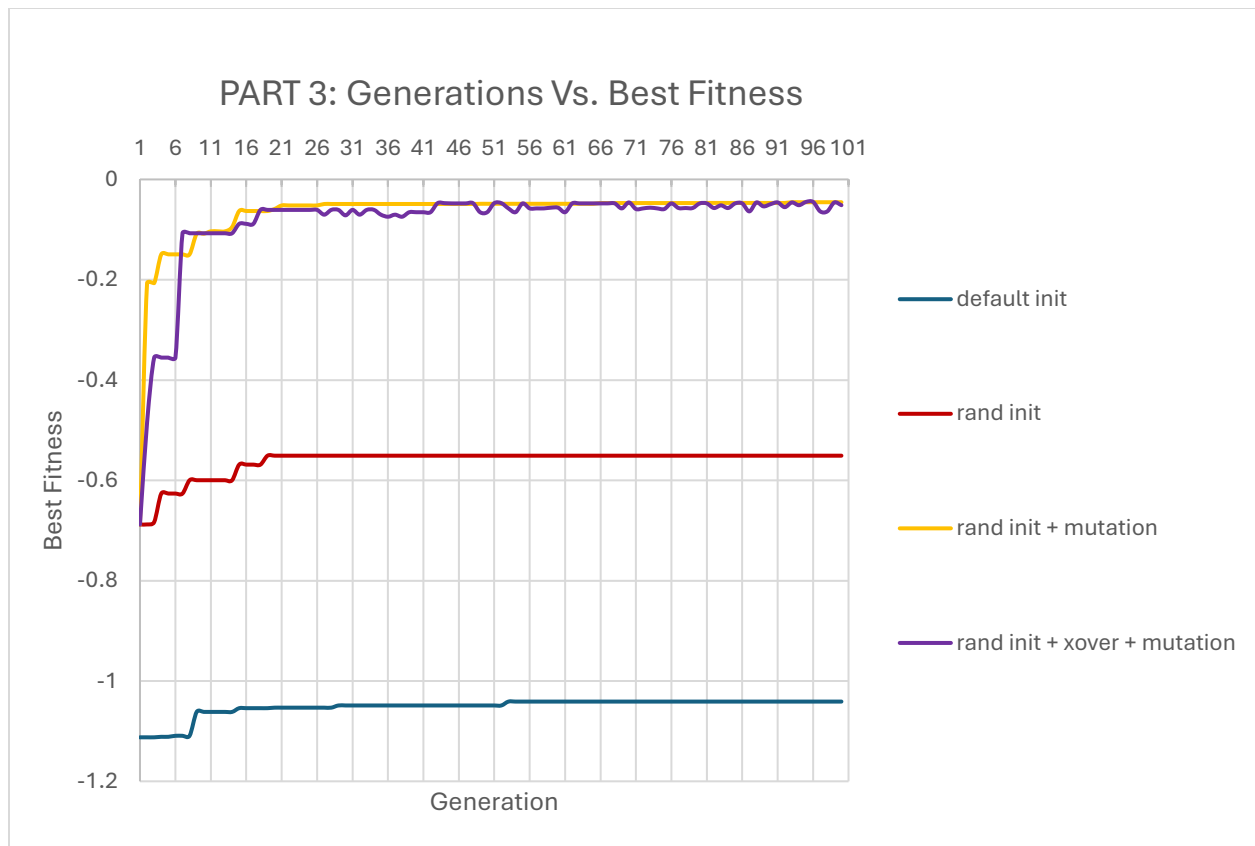


Figure 3: Graph displaying values determined in Part 3 for Genetic Programming.